DSA Practical 3.3 Selection Sort & Insertion Sort

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Batch:A2

Selection Sort

```
#include<iostream>
using namespace std;
float a[100];
int temp,n,min,i,j;
void selection_sort(float a[],int n){
	for (i=0;i<n-1;i++){
		int min=i;
		for(j=i+1;j<n;j++){
		if (a[j]<a[min]){
			min=j;
		}
		temp=a[min];
		a[min]=a[i];
		a[i]=temp;
		}
	}
}

int main(){
	int n;
```

```
	cout<<"Enter Number of Employee: ";
	cin>>n;
	float a[n];
	for(i=0;i<n;i++){
		cout<<"\nEnter the Salary of  "<<i+1<<" Employee: ";
		cin>>a[i];
	}
	selection_sort(a,n);
	cout<<"\nTop five highest salaries in the company: ";
	for(i=0;i<n;i++){
		cout<<a[i]<<"\t";
	}
	return 0;
}

/*
```

Output

```
Enter Number of Employee: 5

Enter the Salary of  1 Employee: 4576

Enter the Salary of  2 Employee: 12453.3
```

Enter the Salary of 3 Employee: 12453.96

Enter the Salary of 4 Employee: 151423.96

Enter the Salary of 5 Employee: 741258.0

Top five highest salaries in the company: 4576 12453 12453
151423 741258

Process exited after 27.38 seconds with return value 0

Press any key to continue . . .

*/

Insertion Sort

```
#include<iostream>
using namespace std;
float a[100];
int temp,n,i,j;
void insertion_sort(float a[],int n){
	for (i=0;i<n;i++){
		temp=a[i];
		j=i-1;
		while(j>=0 && a[j]>temp){
			a[j+1]=a[j];
			j--;
		}
		a[j+1]=temp;
	}
}

int main(){
	int n;
	cout<<"Enter Number of Employee: ";
	cin>>n;
```

```
        float a[n];
        for(i=0;i<n;i++){
                cout<<"\nEnter the Salary of  "<<i+1<<" Employee: ";
                cin>>a[i];
        }
        insertion_sort(a,n);
        cout<<"\nTop five highest salaries in the company: ";
        for(i=0;i<n;i++){
                cout<<a[i]<<"\t";
        }
        return 0;
}

/*

Output

Enter Number of Employee: 5

Enter the Salary of  1 Employee: 5412.36

Enter the Salary of  2 Employee: 85740.36

Enter the Salary of  3 Employee: 4568.21
```

Enter the Salary of 4 Employee: 41258.02

Enter the Salary of 5 Employee: 74125.96

Top five highest salaries in the company: 4568 5412 41258 74125 85740

*/